

🎓 EduGenie – Gemini Learning Assistant

EduGenie is an AI-powered learning assistant designed to help students learn, understand, and revise academic topics using **Google Gemini Generative AI**.

The application can be developed as an interactive learning platform where students enter a topic or question and receive simple, personalized explanations, examples, summaries, quizzes, and study support.

> **Repository:** <https://github.com/priya12pri2007-cpu/EduGenie-Gemini-Learning-Assistant>

📌 Project Overview

EduGenie aims to make learning easier by combining a simple student-friendly interface with Generative AI.

Main Objectives

- Provide AI-based learning assistance.
- Explain difficult concepts in simple language.
- Generate topic summaries and study notes.
- Create practice questions and quizzes.
- Give examples based on the student's question.
- Provide an interactive learning experience.
- Reduce the time required to prepare study material.

✨ Key Features

- 🤖 **AI Learning Assistant**
- 💬 **Question & Answer Support**
- 📖 **Topic Explanation**
- 📝 **Automatic Notes/Summary Generation**
- ❓ **Quiz / Practice Question Generation**
- 🎯 **Personalized Learning Prompts**
- 🧠 **Google Gemini Integration**
- 🌐 **Simple Web-Based Interface**
- ⚡ **Real-Time AI Responses**

🏗️ Project Structure

A suitable full-stack structure for EduGenie is:

```
```text
EduGenie-Gemini-Learning-Assistant/
|
|— app.py
|— requirements.txt
|— README.md
|— .env
|— .gitignore
|
|— templates/
| |— index.html
|
|— static/
| |— css/
| | |— style.css
| |— js/
| | |— script.js
|
|— assets/
| |— images/
|
...
```
```

> **Important:** The GitHub repository currently visible from the supplied link contains project-document PDFs (for example, brainstorming, problem-statement, and empathy-map documents). The source-code files above describe the recommended implementation structure; they should be changed to match the actual code files when the application source is uploaded.

Front-End

The front-end is the part of EduGenie that students directly interact with.

Front-End Technologies

- HTML5
- CSS3
- JavaScript
- Jinja2 templates (if Flask is used)

Front-End Functions

The interface can provide:

1. Student question/topic input.
2. Learning mode selection.
3. Generate button.
4. AI answer display.
5. Summary/notes display.
6. Quiz generation.
7. Clear/reset controls.
8. Responsive student-friendly design.

Example Front-End Flow

```
```text
Student
↓
Enter Topic / Question
↓
Click Generate
↓
JavaScript sends request
↓
Backend API
↓
Gemini AI
↓
AI Response
↓
Display Result on Web Page
```
```

Back-End

The back-end manages the application logic and communication with Gemini AI.

Back-End Technologies

- Python

- Flask
- Google Gemini API / Google Generative AI SDK
- Python-dotenv
- JSON
- HTTP/API handling

Back-End Responsibilities

1. Receive the student's input.
2. Validate the input.
3. Build an appropriate AI prompt.
4. Send the prompt to Google Gemini.
5. Receive the AI-generated response.
6. Process the response.
7. Send the result back to the front-end.
8. Handle API errors safely.

AI Technology Used

Google Gemini Generative AI

The main AI technology used for EduGenie is **Google Gemini**.

Gemini can understand natural-language questions and generate educational responses based on the student's prompt.

AI Capabilities

EduGenie can use Gemini for:

- Concept explanations
- Summaries
- Notes
- Examples
- Practice questions
- MCQs
- Revision assistance
- Personalized learning responses

AI Workflow

```
```text
Student Input
↓
Python Backend
↓
Prompt Creation
↓
Google Gemini API
↓
Gemini Model
↓
Generated Educational Content
↓
Backend
↓
Frontend
↓
Student
```
```

🧠 Prompt Engineering

EduGenie can create structured prompts so that Gemini produces student-friendly answers.

Example Prompt

```
```text
You are EduGenie, an educational AI assistant.
```

Explain the following topic in simple language.

Topic:  
{student\_topic}

Requirements:

- Give a clear definition.
- Explain the concept step by step.
- Give a simple example.
- Mention important points.
- Keep the explanation suitable for a student.

```
```
```

For quiz generation:

```
```text
```

Create 5 multiple-choice questions about:

{topic}

For each question:

1. Give four options.
2. Identify the correct answer.
3. Give a short explanation.

```
```
```

📁 File Description

| File / Folder | Purpose |
|---------------|---------|
|---------------|---------|

| | |
|-----|-----|
| --- | --- |
|-----|-----|

| | |
|----------|---|
| `app.py` | Main Python application and backend logic |
|----------|---|

| | |
|--------------------|---|
| `requirements.txt` | Python packages required by the project |
|--------------------|---|

| | |
|--------|-----------------------------------|
| `.env` | Stores the Gemini API key locally |
|--------|-----------------------------------|

| | |
|--------------|---|
| `.gitignore` | Prevents secret and unnecessary files from being uploaded |
|--------------|---|

| | |
|------------------------|--------------------|
| `templates/index.html` | Main web interface |
|------------------------|--------------------|

| | |
|------------------------|-------------------|
| `static/css/style.css` | Front-end styling |
|------------------------|-------------------|

| | |
|-----------------------|--|
| `static/js/script.js` | Front-end interaction and API requests |
|-----------------------|--|

| | |
|-----------|-------------------------------|
| `assets/` | Images and other UI resources |
|-----------|-------------------------------|

| | |
|-------------|-----------------------|
| `README.md` | Project documentation |
|-------------|-----------------------|

🛠️ Technologies at a Glance

| Category | Technology |
|----------|------------|
|----------|------------|

| | |
|-----|-----|
| --- | --- |
|-----|-----|

| | |
|----------------------|--------|
| Programming Language | Python |
|----------------------|--------|

| | |
|-----------|-----------------------|
| Front-End | HTML, CSS, JavaScript |
|-----------|-----------------------|

| | |
|-------------------|-------|
| Backend Framework | Flask |
|-------------------|-------|

| | |
|----|-----------------------------|
| AI | Google Gemini Generative AI |
|----|-----------------------------|

| | |
|-------------------|-------------|
| API Communication | HTTP / JSON |
|-------------------|-------------|

| | |
|-----------------------|---------------|
| Environment Variables | python-dotenv |
|-----------------------|---------------|

| | |
|-----------------|--------------|
| Version Control | Git & GitHub |
|-----------------|--------------|

🗝️ API Key Configuration

Create a `.env` file in the project folder:

```
```text
GEMINI_API_KEY=your_gemini_api_key
```
```

Do **not** upload the real API key to GitHub.

Add the following to `.gitignore`:

```
```text
.env
__pycache__/
*.pyc
venv/
```
```

📦 Installation

1. Clone the Repository

```
```bash
git clone https://github.com/priya12pri2007-cpu/EduGenie-Gemini-Learning-Assistant.git
```
```

2. Open the Project

```
```bash
cd EduGenie-Gemini-Learning-Assistant
```
```

3. Create a Virtual Environment

```
```bash
python -m venv venv
```
```

Windows

```
``bash
venv\Scripts\activate
``
```

Linux / macOS

```
``bash
source venv/bin/activate
``
```

4. Install Dependencies

```
``bash
pip install -r requirements.txt
``
```

5. Add Gemini API Key

Create `.env`:

```
``text
GEMINI_API_KEY=your_gemini_api_key
``
```

6. Run the Application

For a Flask application:

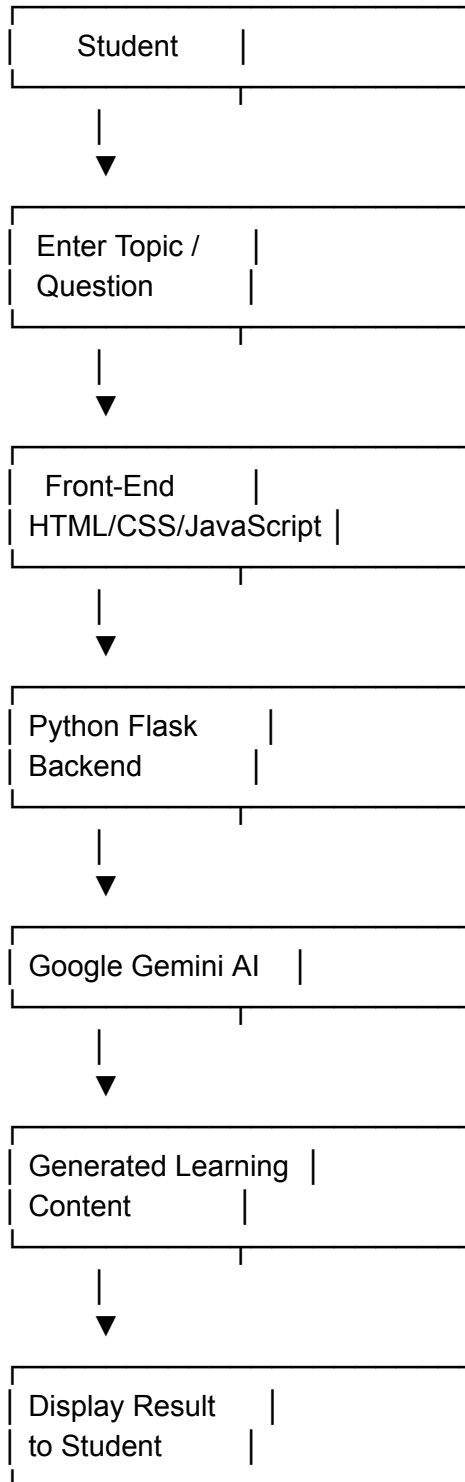
```
``bash
python app.py
``
```

Then open:

```
``text
http://127.0.0.1:5000
``
```

Application Workflow

```text



```

🌟 Innovative Aspects

1. AI-Based Personalized Learning

Instead of showing the same fixed content to every student, Gemini can generate responses based on the student's question and learning requirement.

2. Simple-Language Explanations

Students can request difficult concepts to be explained in simple language.

3. Multiple Learning Modes

The same topic can be converted into:

```
```text
Topic
↓
Explanation
↓
Summary
↓
Examples
↓
Quiz
↓
Revision
```
```

4. Interactive Learning

Students can continue asking follow-up questions instead of only reading static notes.

5. Generative AI Integration

The project demonstrates how a Generative AI model can be integrated into a real student-oriented application.

🎯 Use Cases

EduGenie can be useful for:

- College students
- School students
- Exam preparation
- Quick revision
- Concept clarification
- Practice questions
- Self-learning
- Academic project demonstrations

🌐 Future Enhancements

- 📄 PDF upload and document-based learning
- 🎤 Voice-based questions
- 🗣️ Text-to-speech answers
- 🌐 Multi-language learning
- 📊 Student progress dashboard
- 🧠 Personalized learning paths
- 📝 Automatic question-paper generation
- 🗂️ Flashcard generation
- 📅 AI study planner
- 💾 Learning history
- 🔒 Student login and profiles

⚠️ Security

Never expose your Gemini API key in source code or commit it to GitHub.

Use:

```
```text
.env
```
```

and keep it inside `.gitignore`.

📜 License

This project can be released under the MIT License if the project owner chooses to use that license.

🏠 Project

****Project Name:** EduGenie – Gemini Learning Assistant**

****Repository:****

<https://github.com/priya12pri2007-cpu/EduGenie-Gemini-Learning-Assistant>

****AI Technology:** Google Gemini Generative AI**

****Purpose:** AI-powered educational assistance and personalized learning**